

KYIV, Ukraine

WMO: 333450

Lat: 50.3917N

Lon: 30.5356E

Elev: 167

StdP: 99.33

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-14.1	-20.8	0.6	-15.8	-17.9	0.8	-13.6	10.1	0.2	9.1	-3.1	2.9	30	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.2	20.3	29.6	19.8	28.0	19.0	21.7	28.2	20.9	27.1	20.1	26.1	3.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.8	24.4	18.9	14.0	23.5	18.0	13.2	22.9	64.2	28.5	61.3	27.1	58.6	26.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.5	-20.1	33.6	3.5	2.1	-22.5	35.1	-24.6	36.3	-26.5	37.5	-29.0	39.0		
(5)			-20.4	23.5	3.4	1.0	-22.8	24.2	-24.8	24.8	-26.8	25.4	-29.3	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.7	-2.5	2.3	10.0	15.7	19.2	21.2	20.2	14.7	8.4	2.9	-1.9	(6)
(7)		DBStd	9.93	5.45	5.60	4.51	4.23	4.23	3.48	3.07	3.47	3.86	4.25	4.73	5.27	(7)
(8)		HDD10.0	1746	426	351	241	51	6	0	0	0	6	82	216	368	(8)
(9)		HDD18.3	3696	684	584	498	251	102	32	9	19	119	309	462	626	(9)
(10)		CDD10.0	1357	0	0	1	52	184	275	346	316	148	32	3	0	(10)
(11)		CDD18.3	266	0	0	0	1	22	57	97	77	11	1	0	0	(11)
(12)		CDH23.3	2259	0	0	0	12	190	466	794	691	105	2	0	0	(12)
(13)		CDH26.7	644	0	0	0	2	35	113	241	234	19	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.5	3.4	3.6	3.2	3.0	2.9	2.7	2.6	2.7	2.8	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	631	45	41	39	44	58	73	78	62	53	41	50	50	(15)
(16)		PrecMax	900	149	121	113	132	172	190	216	164	211	144	142	133	(16)
(17)		PrecMin	382	8	4	3	2	12	7	6	3	2	0	5	3	(17)
(18)		PrecStd	112	27	25	24	30	33	41	43	34	43	29	27	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.0	16.4	25.0	29.9	31.2	33.1	34.6	29.0	22.9	16.0	10.1	(19)
(20)			MCWB	5.4	6.6	9.5	15.1	18.5	20.0	21.5	20.6	18.1	15.2	11.9	8.7	(20)
(21)		2%	DB	5.6	7.2	13.8	21.7	27.2	29.4	31.1	31.8	26.1	19.9	13.0	7.5	(21)
(22)			MCWB	4.2	4.9	8.1	12.8	17.2	19.4	20.8	20.4	16.8	14.6	10.8	6.4	(22)
(23)		5%	DB	4.0	5.4	11.1	19.2	25.4	27.9	29.2	29.2	24.0	17.1	11.0	6.0	(23)
(24)			MCWB	2.9	3.7	6.7	11.8	16.7	19.1	20.5	19.6	16.2	13.0	9.3	5.3	(24)
(25)		10%	DB	2.6	3.9	8.9	17.2	23.5	26.1	27.8	27.1	21.5	14.9	9.2	4.3	(25)
(26)			MCWB	1.7	2.4	5.6	10.6	15.8	18.4	19.7	18.7	15.3	11.9	8.0	3.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.2	10.4	15.9	20.0	22.0	23.1	22.9	19.8	17.0	12.9	8.9	(27)
(28)			MCDWB	6.7	9.4	15.3	23.4	26.7	27.8	30.0	30.6	26.7	20.9	14.9	9.9	(28)
(29)		2%	WB	4.3	5.2	8.6	14.0	18.7	20.9	22.1	21.4	18.1	15.1	11.1	6.5	(29)
(30)			MCDWB	5.3	6.7	12.4	19.5	24.8	26.9	28.5	28.2	23.8	18.5	12.6	7.1	(30)
(31)		5%	WB	3.0	3.9	7.2	12.7	17.6	20.0	21.2	20.4	16.9	13.6	9.6	5.2	(31)
(32)			MCDWB	3.8	5.3	10.2	17.5	23.4	25.9	27.3	27.0	22.1	16.7	10.8	5.8	(32)
(33)		10%	WB	1.8	2.6	5.8	11.6	16.6	19.2	20.4	19.6	15.9	12.1	8.1	3.6	(33)
(34)			MCDWB	2.4	3.5	8.5	16.3	22.1	24.7	26.2	25.8	20.5	14.6	9.1	4.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	7.1	9.2	10.2	9.8	9.9	10.2	9.3	7.9	4.8	4.3	(35)
(36)			MCDBR	4.7	6.8	11.2	12.5	12.7	12.2	12.3	13.5	13.5	11.8	7.1	5.5	(36)
(37)			MCWBR	4.0	5.0	6.8	6.2	5.5	5.3	5.0	5.3	6.1	7.1	5.2	4.5	(37)
(38)		5% WB	MCDWR	4.4	6.1	10.0	10.8	11.3	10.8	10.9	11.4	11.3	10.5	6.1	5.2	(38)
(39)			MCWBR	3.9	4.9	6.6	5.9	5.3	5.3	4.9	5.1	5.8	6.8	5.0	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.337	0.368	0.432	0.406	0.398	0.427	0.429	0.397	0.357	0.339	0.308	(40)	
(41)		taud	2.226	2.176	2.233	2.137	2.289	2.330	2.261	2.242	2.296	2.385	2.324	2.267	(41)	
(42)		Ebn at Noon	705	776	825	808	852	860	826	803	788	756	666	646	(42)	
(43)		Edh at Noon	73	99	113	139	125	121	128	124	106	81	67	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.86	1.69	2.98	4.34	5.68	6.00	5.83	5.07	3.46	1.98	0.79	0.58	(44)	
(45)		RadStd	0.15	0.29	0.47	0.44	0.45	0.44	0.38	0.31	0.51	0.32	0.12	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.62	N/A	N/A	+1.08	N/A	N/A	N/A	-130	+76	N/A	
(47) Regional (10 neighbors)	+0.74	N/A	N/A	+1.00	+0.50	N/A	-140	-201	+120	+55	

Nomenclature: See separate page